Money Mate – Personal Finance & Peer-to-Peer Loan Tracker

---

1. Problem Statement

Managing personal finances is a challenge for many individuals, especially when it comes to tracking income, expenses, budgeting, and managing interpersonal loans. Traditional financial tracking tools often lack personalization and flexibility. They also don't offer a way to manage money lent to or borrowed from friends, which can lead to misunderstandings and lack of accountability. Moreover, existing solutions are often cluttered or inaccessible to everyday users who just want a clear picture of their financial health.

---

2. Solution Statement

Introducing **Money Mate**, a user-friendly, secure, and affordable platform that helps users take control of their personal finances. With features like income/expense tracking, loan management, budgeting tools, and peer-to-peer financial accountability, Money Mate empowers individuals to make better financial decisions. It also enables users to record money lent to or borrowed from friends and track installments—bringing structure to casual financial exchanges. Insightful analytics, budget alerts, and mobile-friendly design ensure users stay informed and on track anytime, anywhere.

---

3. Core Features

User System

- User registration and login using JWT authentication
- User roles: Regular User (default)
- Profile management: name, email, profile photo, and preferred currency
- Dashboard showing income/expense summary, active loans, and financial goals

---

Income & Expense Tracking

- Add, edit, delete income entries with amount, source, date, and notes
- Add, edit, delete expense entries with category, amount, date, and notes
- Categorize expenses (e.g., food, transport, entertainment, etc.)
- View monthly summaries of income and expenses

Loan Management

- Record loans: lent or borrowed, with amount, parties involved, and purpose
- Set installment plans with due dates
- Track loan status: active, paid, overdue
- Reminders for upcoming due payments
- Record partial payments and view outstanding amounts

Budget Planner

- Set monthly or yearly budgets
- Get alerts when nearing or exceeding limits
- Visual display of remaining budget in each category

Insights and Analytics

- Bar graphs, pie charts, and trend lines for financial data
- Analyze income/expense trends over time
- View savings, debt status, and cash flow at a glance

Communication

- In-app notifications for loan due dates, budget alerts, and milestone achievements
- Email notifications for reminders and payment confirmations

---

Community Engagement

- **User Blogs**: Users can write and publish blogs about personal finance, savings, budgeting tips, etc.
- **Likes**: Other users can like blogs to show appreciation (one like per user per blog).
- **Comments**: Logged-in users can comment on blogs; authors and commenters can edit/delete their own comments.
- **Sorting & Filtering**: Blogs can be sorted by recent, most liked, or most commented. Tags and search available.
- **Profile Integration**: Blog and comment history is visible on user profiles with like counts and engagement metrics.

---

4. Advanced Features

Availability and Scheduling

- Calendar view of upcoming loan due dates, payments, and budget deadlines
- Option to export schedule to Google Calendar (future integration)

Search and Recommendation System

- Smart search for previously entered transactions, people, or categories
- Suggestions for categorizing expenses based on past entries

Frontend & Backend (Unified)

- **Next.js** – Full-stack framework for UI, APIs, and server-side rendering
- **Tailwind CSS** – For styling
- **API Routes** – To implement RESTful endpoints inside `pages/api/`

Database

- **MySQL** (via Prisma or Sequelize)

Authentication

- **JWT**, and optionally **OAuth (e.g., Google)**

Notifications

- **SendGrid** for email
- **In-app alerts** using client state or sockets

Integrations

- **Stripe** or **Razorpay** for payments
- **Google Calendar API** for scheduling

---

7. Non-Functional Requirements

Performance

- Must support up to 10,000 active users concurrently
- API response time < 500ms for standard operations

Security

- Data encryption for sensitive information
- Secure HTTPs connections
- Authentication tokens with refresh system

Reliability

- 99.5% uptime per month
- Daily database backups

Usability

- Minimalist UI with user-friendly navigation
- Mobile-first responsive design
